



GÜRKAN DOĞUKAN KARAKURT

MSC CANDIDATE & RESEARCH ASSISTANT

SUMMARY

MsC candidate and Research Assistant in Construction Engineering and Management at Middle East Technical University. Building on my foundation as a Civil Engineer, my core expertise lies at the intersection of Artificial Intelligence and the built environment. I actively research and develop advanced machine learning models to predict and automate construction schedules directly from BIM topology. Previously, as a Software Engineer at Mega Mühendislik, I built custom automation tools and software to optimize civil engineering operations and drive digital transformation in construction workflows. Beyond my primary academic and engineering focus, I possess a versatile background as a full stack developer and designer. I have briefly contributed to a select number of external software initiatives, ranging from quantitative algorithmic trading systems at Vortex to serving as the technical lead for an AI powered wellness application. Ultimately, I consistently bridge the gap between complex civil engineering research and practical application by engineering scalable software architectures.

EDUCATION

2025-Ongoing	MASTER'S IN CONSTRUCTION MANAGEMENT Orta Doğu Teknik Üniversitesi (METU)
2017-2023	CIVIL ENGINEERING Orta Doğu Teknik Üniversitesi (METU)/ 3.35 GPA
2019-2023	CORPORATE FINANCE (MINOR PROGRAM) Orta Doğu Teknik Üniversitesi (METU)/ 3.39 GPA

EXPERIENCE

Research Assistant Middle East Technical University October 2025 to Present (Ankara)

I conduct graduate research in Construction Engineering and Management under the supervision of Prof. Dr. Rifat Sönmez. My research focuses on developing machine learning models to predict and automate construction schedules directly from BIM topology. Additionally, I serve as a teaching assistant for departmental courses, where I provide academic support, prepare course materials, and facilitate student learning within the civil engineering program.

Software Engineer (Construction Tech.) Mega Mühendislik Müşavirlik A.Ş. Dec 2024 - Oct 2025 (Ankara)

Working on the digital transformation of construction workflows by building internal automation tools and custom software for engineering operations. Focused on data structuring, process optimization, and technical problem-solving for civil engineering applications.

Lead Frontend Designer Global Crypto Exchange (Stealth)

Jan 2025 - Aug 2025 (Remote)
Spearheading the design and implementation of the core trading interface for a pre-launch global crypto exchange. Responsible for responsive architecture, performance optimization, and UX/UI systems that adapt across devices. Technologies include React, TypeScript, Tailwind, and WebSocket-based live data feeds.

Head of Quantitative Strategies Vortex Foundation 2023 - 2024 (Remote)

Led the design, development, and deployment of algorithmic trading bots for hedge fund and retail-level use cases. Delivered strategies based on user-specific risk appetites, including scalping, swing, and long-term investment models. Built fully autonomous bots with live market execution and order book integration. Also developed custom market making algorithms for liquidity optimization on multiple exchanges.

Co Founder & Full Stack Developer Diyet Cebimde

2023 to Present (Remote)
I architected and developed an AI assisted wellness and diet generation platform in collaboration with my wife, Dietitian Melisa Karakurt. Designed specifically to digitize and scale her professional dietary practice, I built the entire technical infrastructure from the ground up featuring both web and mobile clients. I engineered a hybrid pipeline that integrates the Gemini API for intelligent meal selection and PuLP linear programming for precise portion optimization. The robust backend architecture is built using Python, Django REST Framework, and PostgreSQL, while the admin panel and cross platform mobile client were designed using React 18 and React Native with Expo.

ABILITIES

Software Developer

- Python&Django
- Web Dev(HTML,CSS,JS)
- React, Tailwind
- React Native (Mobile)
- PostgreSQL
- Matlab, VBA
- Algorithmic Trading Development

Civil Engineer

- Construction Software Automation
- Autocad
- Excel-Visual Basics
- Revit
- Navisworks
- MS Project

LANGUAGES

Turkish

English (C1)

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